

Nathan Joseph Svoboda

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Education

Bachelor of Science, Computer Science at *Barrett the Honors College at Arizona State University* August 2022 - May 2026
- Awards: Dean's List 5 Semesters (3.5 GPA or above)

Technical Skills

Programming Languages: Python, C, JavaScript, HTML, CSS, C++, Java, SQL, Bash
Frameworks: React, Pytorch, GDB, Gradle, Baja, Niagara, Postman, Pandas, Numpy, Djitellopy, FastAPI
Project Management: Git, Scrum, ClickUp

Professional Experience

Full-Stack Software Engineering Intern - Hawkeye Energy Solutions May 2024 - Present

- Designed new React components for a customer-facing front-end application by leveraging state management, HTML, and CSS, allowing customers to set maximum and minimum settings through a user-friendly design
- Redesigned the backend of a React application to fix API call and data resolution errors by fixing Java methods that would communicate directly with building meters through integrated servers
- Spearheaded the migration and integration of historical metering data from an older database to a newer and more dynamic database using SQL to provide building metering trends from the past for a client
- Currently working in a CI/CD environment for multiple customer-facing applications utilizing npm, Gradle, GitHub, Java, and the Niagara framework to maintain and develop applications and documentation for customers

Software Engineer - Paragon Autonomous January 2024 - May 2025

- Built software for an autonomous fire surveillance drone using Python and TensorFlow, allowing it to fly with manual controls, notify the host computer when prompted, and visualize its position using the Pygame library
- Implemented A* algorithm to find the best path using point clouds in 3D space and with computation restrictions using C++ to provide autonomous movement and navigate to the site of a potential fire hazard when found
- Researched and developed a template framework in C++ to simplify code deployment with flight controllers, and ensured compatibility with Mavros, Gazebo, and Ardupilot for drone programming and functionality
- Tested the Lidar and Ultrasonic sensors to see the information that is provided, and how that data can be integrated into our code for adaptation of flight controls, and collision avoidance

Projects

Barrett Thesis Project - AI voice clone | *FastAPI, REST API's, Telephony, VAPI, Pytorch* August 2025 - May 2026

- Designed and implemented an AI voice cloning and telephony workflow inside MyShort.biz using React and FastAPI, which includes agent configuration, phone number routing, testing, and live browser conversation
- Built the backend voice pipeline by integrating a local voice cloning path with PyTorch and TorchAudio to generate WAV previews and telephony PCM audio, while tracking call session state with SQL and improving reliability through background jobs, cache warmup, file cleanup, and health checks
- Engineered the business phone agent through Vapi by creating accurate assistants based on a business description, and custom chat webhook endpoints, gaining experience with Python speech services and webhook security

Personal Website | React, HTML, CSS, Javascript, SQL August 2024 - Present

- Built a responsive personal portfolio using React, JavaScript, HTML, and CSS and a D3 visualization to demonstrate Dijkstra's and Prim's algorithms, allowing visitors to explore shortest-path behavior
- Integrated EmailJS contact form and optional Google OAuth hook, managed builds with npm and published via GitHub Pages to streamline deployment and ensure accessibility for various browsers and browser sizes
- Provisioned and configured a MongoDB database and implemented connection logic for backend storage

AI Citizen Science Mobile App | TypeScript, React Native, Expo, Python/FastAPI, GitHub January 2026 - May 2026

- Built full-stack mobile prototype with responsive navigation, camera image capture and upload, AI assisted workflow, and REST API endpoints for health checks, heatmap data, and demo AI responses.
- Developed biodiversity heatmap analytics with React Native Maps, timeline filtering, hotspot aggregation, fallback loading, and pytest validation over 48 Indonesia sightings across 6 categories.